

Dataset link - <https://www.kaggle.com/datasets/farzadnekouei/pothole-image-segmentation-dataset>

Paper link - <https://ieeexplore.ieee.org/document/10542661>

Project Report - Real-Time Pothole Detection & Severity Analysis

Course Code - BCSE403L - Digital Image Processing

Domain - Smart Cities & Transportation Infrastructure

Topic - Real-time Pothole Detection using Hybrid Deep Learning (YOLOv8) and Soft Computing (Fuzzy Logic)

Name - Shrutika

Reg No. – 23BCE0116

1. Introduction

The Problem

Potholes and road defects are a major safety concern worldwide. They are the cause of a significant number of road accidents, severe vehicle damage (tire blowouts, suspension failure), and traffic congestion. Most municipal corporations to date rely on reactive measures to repair roads only after accidents occur or citizens file complaints. Proactive inspection is often manual, slow, and expensive; hence, it is impossible to monitor large road networks effectively in real time.

Domain Selection & Relevance

This particular problem belongs to the category of Smart Cities and Transportation Infrastructure. It is imperative to tackle this particular problem in order to modernize the management of cities. We can easily move from a "reactive" type to a "predictive" type by carrying out the automation of road inspection. A solution based on the current dashcam technology available to us can detect the severity of the potholes in roadways instantly and, in the long run, can save lives and the city's public expenditures on road repairs.

Core Concept

In our proposed solution, a combination of models will be used. Digital Image Processing (DIP) and Deep Learning (YOLOv8) are used to visually detect the pothole, and Soft Computing (Fuzzy Logic) concepts will be used to simulate human thinking to determine the level of damage.

2. Study Section

Base Paper Reference

Title - *POT-YOLO - Real-Time Road Potholes Detection using Edge Segmentation based YOLOv8 Network* **Source** - IEEE Sensors Journal (2024)

Gap Identified - While the base paper achieves high accuracy in *detecting* potholes, it relies on "Crisp Logic" (Binary detection - Yes/No). It lacks a mechanism to prioritize repairs based on severity, treating a minor crack the same as a dangerous crater.

Comparative Analysis of Existing Models

Feature	Traditional IP (Canny/Thresholding)	Deep Learning Only (Base Paper / YOLOv8)	Our Proposed Approach (Hybrid)
Detection Method	Edge detection, Morphological operations.	CNN-based Feature Extraction (Bounding Box/Mask).	Instance Segmentation (YOLOv8) + Fuzzy Logic.
Strengths	Fast; requires low computational power.	High accuracy in detection; robust to lighting changes.	Actionable Intelligence - Provides a "Severity Score" (Low/Med/High).
Limitations	High error rate; confuses shadows/oil spills with potholes.	Binary Output - Only tells you <i>where</i> the hole is, not <i>how bad</i> it is.	Slightly higher computational cost due to the dual-layer processing.
Decision Logic	Hard Computing (If pixel > X, then pothole).	Hard Computing (Confidence Score).	Soft Computing (Fuzzy Inference System).

Dataset & DIP Techniques

- Dataset** - We are utilizing the **Pothole Image Segmentation Dataset** (by Farzad Nekouei). Unlike older datasets that only used bounding boxes, this dataset uses **Polygon Annotations**. This allows our model to learn the exact jagged shape of the pothole, which is crucial for calculating the area.
- DIP Techniques Applied** -
 - Image Resizing** - 640x640 standard input for YOLO.
 - Mosaic Augmentation** - Combining 4 training images into one to help the model learn to detect potholes in complex scenes.
 - HSV Color Space Augmentation** - Adjusting Hue/Saturation to simulate different times of day (morning/evening) and weather conditions.

3. Objectives & Justification

Objective 1 - Precise Instance Segmentation

Statement - To develop a Deep Learning model using YOLOv8 that performs **Instance Segmentation** to extract the exact pixel-level shape (mask) of road defects rather than simple bounding boxes.

Justification - State-of-the-art models mainly utilize Object Detection (Bounding Boxes). A square box is a lousy way to represent a pothole as it includes healthy road sections within the box, which leads to wrong size calculations. Segmentation will allow us to move toward calculating the exact pixel area of the damage. It hence removes the limitation of precision that we have experienced in the Base Paper

and allows accurate raw data to be collected; this raw data is in the form of Area, Width, and Height, which would be required for any type of severity analysis.

Objective 2 - Severity Estimation using Soft Computing (Fuzzy Logic)

Statement - To implement a **Fuzzy Inference System (FIS)** that processes the geometric data from Objective 1 to classify potholes into linguistic severity categories - *Low Priority, Medium Priority, and High Priority*.

Justification - Real-world decision-making is rarely black and white (Hard Computing). A pothole isn't just "present" or "absent"; its danger depends on a combination of its depth, area, and shape.

- **The Gap** - Pure Deep Learning models lack this reasoning capability.
- **The Solution** - By adding a Soft Computing layer, we mimic human judgment. For example, a rule such as *"IF Area is Large AND Shape is Deep THEN Severity is Critical"* handles ambiguity better than rigid mathematical formulas. This makes the system useful for municipal authorities who need a **prioritized list** of repairs, not just a raw map of detections.